

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5.	(a)			(biological) catalysts / Speed up rate of reaction	1			1		
	(b)			Liquid form would be digested in the stomach (1) Would not reach small intestine (1)	1		1	2		
	(c)			Small intestine is alkali (around pancreas) (1) so C is most suitable (1) optimum in higher pH/ pH 8 / pH 7.5 (1) A and B would denature (1) Or Small intestine is alkali (around pancreas) (1) A and B not suitable (not alkaline) (1) C optimum pH is alkali (1) A and B would denature (1)	1	3				
	(d)	(i)		Stains caused by different types of food/food contains different types of molecules	1					
		(ii)		Any 3 × (1) from Enzymes will denature (1) Active site changed/deforms (1) affects lock and key model (1) Bonds broken (1) Prevents enzyme-substrate complex forming (1) + Food stain wouldn't be broken down (1)	4					
				Question 5 total	8	3	1	12		